

Estrella Solitaria in offset curves

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Estrella Solitaria, the *Lone Star Flag*, official flag of Cuba, was designed in 1849, 175 years ago. “It has become much more than a visible symbol for the island nation: Modern states and would-be nations around the globe have imitated the elegant design of this flag”. [1]

Estrella Solitaria (similarly to the Texas Navy Flag, being created already in 1836) contains a regular pentagram that consists of straight segments. Surprisingly, an algebraic curve that recalls the Lone Star Flag, can be created with a simple construction taken from differential geometry, by using a recent version of GeoGebra Discovery*.

We consider the algebraic curve $\mathcal{C}_n : y = x^{2n}$ for $n > 1$ and compute the *offset curve* $\mathcal{O}_{n,d}$ (also known as *parallel curve*) in a distance d from $\mathcal{C}_n$. If d is large enough, $\mathcal{O}_{n,d}$, itself an algebraic curve of degree $8n - 2$, contains cusps that form a pentagram-like shape.

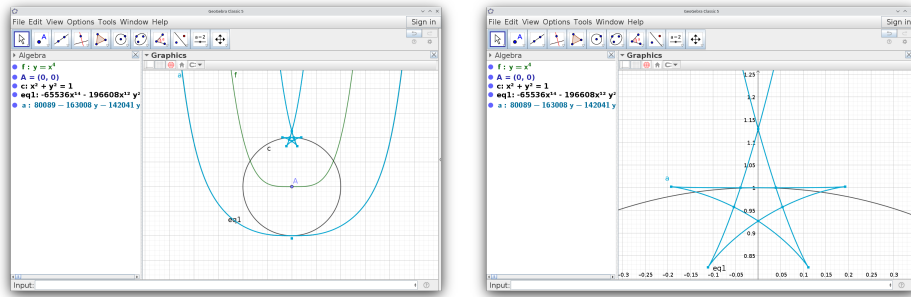


Figure 1: $\mathcal{C}_2$ and $\mathcal{O}_{2,1}$

Figure 1 was created by using GeoGebra Discovery’s **Envelope** tool (or command). In the background it uses a Jacobi matrix, Gröbner bases, elimination, and a fast numerical plotting algorithm to sketch up the output curve (here, degree 14) very quickly [2]. The sketch can be, however, inaccurate, and in many cases important parts of the pentagram form may be missing.

*<https://github.com/kovzol/geogebra/releases>

The way how these offset curves are created is simple [4]:

1. The user creates $f : y = x^{2n}$ by typing its formula in the Input Bar (this plays the role of C_2).
2. A circle c with fixed radius 1 is put with its center A on f .
3. After selecting the **Envelope** tool from the Toolbar, and clicking on c and A , the offset curve $eq1$ is shown (this plays the role of $\mathcal{C}_{2,1}$).

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--prover=timeout:1200,
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In fact, in general there may be difficulties in computing offset curves for arbitrary inputs of algebraic curves. GeoGebra Discovery’s method assumes the input in the form of an implicit equation. If the input curve is of degree higher than 4, the Gröbner basis computation is usually infeasibly slow. By contrast, very interestingly, our setup with the input curves C_n allows remarkably fast symbolic computations, even for $n = 5$, at least for plotting the sketch (see Figure 2).

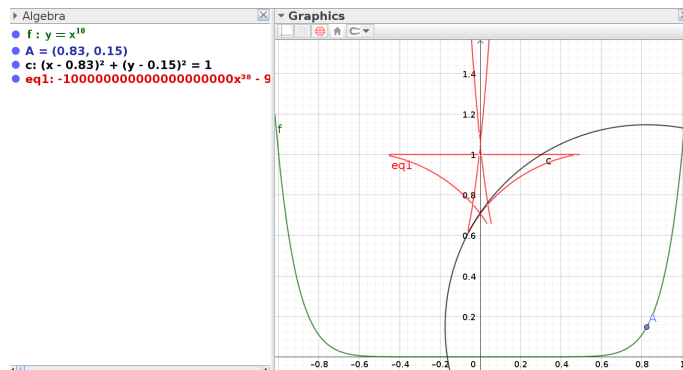


Figure 2: C_5 (degree 10) and an inaccurate sketch of $\mathcal{O}_{5,1}$ (degree 38), computed in 1371 milliseconds on a typical personal computer

The presence of the pentagram-like form can be explained with some background in differential geometry [6,7,8]. As it is known, offset curves will have cusps when the distance from the curve matches the radius of curvature. On the other hand, cusps can be present only if the curvature has a local extremum. For the observed curves $\mathcal{C}_n$, it can be shown that the curvature has two extrema, therefore two cusps appear and they together form a pentagram if d is well-chosen.

Because of its simple creation process and the reasonably fast performance, we find this family of curves a good option to introduce notions from differential geometry in the classroom in a modern way [9].

Example computation

By following the steps 1–3 given above, the algebraic translation of the geometric construction consists of the following computations in the case of $n = 2$ (all computations are created automatically by GeoGebra Discovery, without any user intervention):

1. Circle c is given with center $A = (v_1, v_2)$, auxiliary point $P = (v_3, v_4)$ and radius $v_5 = 1$ where $v_3 - v_1 = v_5$ and $v_4 = v_2$.
2. Another auxiliary point $Q = (v_6, v_7)$ is put on c by using the formula $(v_1 - v_6)^2 + (v_2 - v_7)^2 = (v_1 - v_3)^2 + (v_2 - v_4)^2$.
3. The quartic f is defined by $v_2 = v_1^4$.
4. Now the polynomials

$$\begin{aligned} p_1 &= -v_5 + v_3 - v_1, \\ p_2 &= v_4 - v_2, \\ p_3 &= 1 - v_5, \\ p_4 &= -v_7^2 - v_6^2 + v_4^2 + v_3^2 + 2v_7v_2 - 2v_4v_2 + 2v_6v_1 - 2v_3v_1, \\ p_5 &= v_2 - v_1^4 \end{aligned}$$

are considered, they are 5 polynomials in 7 variables.

5. To reduce the number of variables but keep the further computations still feasible, linear substitutions are performed. First, v_5 is substituted by 1 according to $p_3 = 0$, then v_2 is substituted by v_4 according to $p_2 = 0$, then v_1 is substituted by $v_3 - 1$ according to $p_1 = 0$, in all occurrences in polynomials $p_1, p_2, \dots, p_5$. This will result in polynomials

$$\begin{aligned} q_4 &= -v_3^2 - v_4^2 - v_6^2 - v_7^2 + 2v_3v_6 + 2v_4v_7 + 2v_3 - 2v_6, \\ q_5 &= -v_3^4 + 4v_3^3 - 6v_3^2 + 4v_3 + v_4 - 1, \end{aligned}$$

2 polynomials in 4 variables.

6. Now, the determinant of the Jacobian matrix

$$D = \begin{vmatrix} \frac{\partial q_4}{\partial v_3} & \frac{\partial q_5}{\partial v_3} \\ \frac{\partial q_4}{\partial v_4} & \frac{\partial q_5}{\partial v_4} \end{vmatrix}$$

can be obtained, in our case, $D = 8v_3^3v_4 - 8v_3^3v_7 - 24v_3^2v_4 + 24v_3^2v_7 + 24v_3v_4 - 24v_3v_7 + 2v_3 - 2v_6 - 8v_4 + 8v_7 - 2$.

7. Finally, we eliminate all variables but v_6 and v_7 from the ideal $\langle q_4, q_5, D \rangle$, and, after substituting v_6 by x and v_7 by y we get the output $-65536 x^{14} - 65536 x^8 y^6 - 196608 x^{10} y^4 - 196608 x^{12} y^2 + 458752 x^{12} + 262144 x^6 y^6 + 983040 x^8 y^4 + 1179648 x^{10} y^2 + 131072 x^4 y^7 + 393216 x^6 y^5 + 379392 x^8 y^3 + 172544 x^{10} y - 1376256 x^{10} - 393216 x^4 y^6 - 1966080 x^6 y^4 - 2949120 x^8 y^2 + 786432 x^2 y^7 + 2015232 x^4 y^5 + 1772544 x^6 y^3 + 391680 x^8 y + 2293031 x^8 - 65536 y^8 + 65536 x^2 y^6 + 1797120 x^4 y^4 + 3783680 x^6 y^2 + 131072 y^7 - 1998848 x^2 y^5 - 4257280 x^4 y^3 - 2062848 x^6 y - 2216560 x^6 + 147456 y^6 - 371712 x^2 y^4 - 3233472 x^4 y^2 - 423424 y^5 + 1720832 x^2 y^3 + 2114482 x^4 y + 1826112 x^4 - 19968 y^4 + 481536 x^2 y^2 + 455360 y^3 - 508416 x^2 y - 175360 x^2 - 142041 y^2 - 163008 y + 80089 = 0$, a degree 14 curve in variables x and y .

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Keywords

Differential geometry, Offset curves, Automated reasoning in geometry, GeoGebra, Cylindrical algebraic decomposition

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